Exercise 9.2 SOLUTION

1. **Consider a data set at monthly sales figures. A polynomial regression model $P(x) = x^3 + 2x^2 + x - 3$ is fitted to this data. If the observed sales in the 5th month are 240 units, find the percentage error.**

Sol:

$$P(x) = x^3 + 2x^2 + x - 3$$

$$P(5) = (5)^3 + 2(5)^2 + 5 - 3 = 125 + 2(25) + 5 - 3 = 177$$

$$Error = Observed\ Value - Predicted\ Value = |240 - 177| = 63$$

$$Percentage\ Error = \frac{Absolute\ Error}{Observed\ Value} \times 100\% = \frac{63}{240} \times 100\% = 26.25\%$$

2. **A retailer company has developed a polynomial regression model to predict weekly product demand: $D(w) = w^3 - 2w^2 + 5w - 4$, where $D(w)$ represents predicted demand (in units) and w is the week number. Use the remainder theorem to predict demand for 3rd week. If the observe demand is 22 units, calculate the prediction error.**

Sol: Given the demand function $D(w) = w^3 - 2w^2 + 5w - 4$, for the 3rd week $(w = 3)$:

$$D(3) = (3)^3 - 2(3)^2 + 5(3) - 4$$

$$D(3) = 27 - 18 + 15 - 4 \Rightarrow D(3) = 20$$

Prediction Error:

Observed Demand $= 22\ units$

Predicted Demand $= 20\ units$

Prediction Error=Observed Demand-Prediction Demand

Prediction Error $= 22 - 20 = 2$

So, the predicted demand is $20\ units$, and the prediction error is $2\ units$.

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3. **A digital signal processing system has a transfer function with a numerator $B(z) = z^2 - z - 2$. Use the factor theorem to find the zeros of the system.**

Sol: Given $B(z) = z^2 - z - 2$. We want to find the values of z such that $B(z) = 0$.

$z^2 - z - 2 = 0$

Factoring the quadratic:

$(z-2)(z+1) = 0$

Setting each factor to zero:

$z - 2 = 0 \Rightarrow z = 2$

$z + 1 = 0 \Rightarrow z = -1$

Therefore, the zeros of the system are $z = 2, -1$.

4. **A signal process system has a transfer function $H(z) = \dfrac{z^2 + 3z + 2}{z^2 - 0.2z + 0.9}$. Find the zero(s) of the transfer function by using factor theorem.**

Sol: Given the numerator of the transfer function:

$B(z) = z^2 + 3z + 2$

To find the zeros, we set $B(z) = 0$:

$x^2 + 3z + 2 = 0$

Factoring the quadratic:

$(z+1)(z+2) = 0$

Setting each factor to zero to find the roots (Zeros):

$z + 1 = 0 \Rightarrow z = -1$

$z + 2 = 0 \Rightarrow z = -2$

Therefore, the zeros of the transfer function are $z = -1, -2$

5. **A signal process system has a transfer function $H(z) = \dfrac{z^2 - 0.5z - 0.5}{z^3 + 1}$. Find the zero(s) of the transfer function by using factor theorem.**

Sol: To find the zeros of $H(z) = \dfrac{z^2 - 0.5z - 0.5}{z^3 + 1}$, we need to find the roots of the numerator:

$z^2 - z + 0.5z - 0.5 = 0$

$z(z-1) + 0.5(z-1) = 0$

$(z-1)(z+0.5) = 0$

$z - 1 = 0$ *or* $z + 0.5 = 0$

$z = 1$ *or* $z = -0.5$

6. **A signal processing system has a transfer function with a denominator $A(z) = z^2 - 0.3z - 0.4$. Use factor theorem to find the poles of the system and determine if the system is stable.**

Sol: Given the denominator $A(z) = z^2 - 0.3z - 0.4$

To find the poles, we set $A(z) = 0$:

$z^2 - 0.3z - 0.4 = 0 \Rightarrow z^2 - 0.8z + 0.5z - 0.4 = 0$

$z(z - 0.8) + 0.5(z - 0.8) = 0$

$(z + 0.5)(z - 0.8) = 0$

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Setting each factor to zero to find the poles:

$z - 0.8 = 0 \quad \Rightarrow \quad z = 0.8$

$z + 0.5 = 0 \quad \Rightarrow \quad z = -0.5$

The poles of the system are located at $z = 0.8$ and $z = -0.5$

Stability Assessment:

For stability, the magnitude of each pole must be less than 1:

$|0.8| = 0.8 < 1$

$|-0.5| = 0.5 < 1$

Since both poles have a magnitude less than 1, the system is stable

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7. The denominator of signal processing system's transfer function equal to $A(z) = z^2 + 1.2z + 0.35$. Use factor theorem to determine the location of the corresponding poles and assess the stability of the system.

Sol: Given the denominator $A(z) = z^2 + 1.2z + 0.35$

To find the poles, we set $A(z) = 0$:

$z^2 + 1.2z + 0.35 = 0$

$\Rightarrow z^2 + 0.7z + 0.5z + 0.35 = 0$

$\Rightarrow z(z + 0.7) + 0.5(z + 0.7) = 0$

Factoring the quadratic:

$\Rightarrow (z + 0.5)(z + 0.7) = 0$

Setting each factor to zero to find the poles:

$z + 0.5 = 0 \quad \Rightarrow \quad z = -0.5$

$z + 0.7 = 0 \quad \Rightarrow \quad z = -0.7$

The poles of the system are located at $z = -0.5$ and $z = -0.7$

Stability Assessment:

For stability, the magnitude of each pole must be less than 1:

$|-0.5| = 0.5 < 1$

$|-0.7| = 0.7 < 1$

Since both poles have a magnitude less than 1, the system is stable.

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